

Problem-Solution Fit

Date	4 July 2026
Team ID	XXXXXX
Project Name	Human Development Index (HDI) Prediction Using Machine Learning
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT * Government Policy Makers * Researchers * Data Analysts * Development Organizations	5. CUSTOMER LIMITATIONS * Limited technical knowledge * Limited access to advanced analytical tools * Time constraints * Budget limitations	8. AVAILABLE SOLUTIONS PROS & CONS [AS] Available Solutions • Manual HDI calculations • Spreadsheet-based analysis • Statistical reports Pros • Simple to understand • Widely available Cons • Time-consuming • Prone to human errors • No instant prediction
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] * Predicting HDI manually is time-consuming. * Multiple indicators must be analyzed together. * High chance of calculation errors. * Frequently needed for development planning and analysis.	6. PROBLEM ROOT / CAUSE [RC] * HDI depends on multiple indicators. * Manual calculations are complex. * Lack of automated prediction systems. * Data analysis requires expertise.	9. BEHAVIOR + ITS INTENSITY [BE] * Frequently analyzes development data. * Looks for accurate and quick results. * Uses online tools whenever available.

3. TRIGGERS TO ACT [TR]

- * Need to evaluate a country's development level.
- * Compare HDI across different countries.
- * Support policy and planning decisions.
- * Obtain quick and accurate predictions.

4. EMOTIONS BEFORE / AFTER [EM]

Before

- Confused
- Frustrated
- Unsure about accuracy

After

- Confident
- Satisfied
- Better informed for decision-making

7. YOUR SOLUTION [SL]

Develop a Machine Learning-based HDI Prediction System using Linear Regression that predicts the Human Development Index from Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income Per Capita. The solution is deployed as a Flask web application, enabling users to enter values and instantly receive the predicted HDI score and HDI category.

10. CHANNELS OF BEHAVIOR [CH]

ONLINE

- * Government websites
- * UNDP reports
- * HDI prediction web application

OFFLINE

- * Research papers
- * Government reports
- * Policy meetings